

Reduction of Magellan/LDSS3-C spectroscopy with `PYPEIT`: pipeline description and commissioning results

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Abstract

We describe the support for the Low Dispersion Survey Spectrograph 3 (LDSS3-C) on the Magellan Clay telescope that we have implemented in the open-source spectroscopic pipeline `PYPEIT`, and we report its performance on archival Las Campanas Observatory commissioning data and on science data taken between 2018 and 2025. The three volume-phase holographic grisms (VPH-Blue, VPH-Red and VPH-All) are supported in longslit and multi-slit mode. The pipeline assembles the two amplifier files that LDSS3-C writes for each exposure, classifies frames from the header, traces slits, builds flat fields, calibrates the wavelength scale against archived arc-lamp templates, and performs two-dimensional sky subtraction and optimal extraction. From bias frames at 5 epochs we measure a read noise of 6.44 ± 0.11 and 6.85 ± 0.10 e^- for the two amplifiers in the Fast readout mode, consistent with the published values and 35–38 % larger than the values recorded in the `ENOISE` header keyword. Arc-lamp wavelength solutions of the 1 arcsec slits reach median RMS residuals of 0.04, 0.15 and 0.23 Å for VPH-Blue, VPH-Red and VPH-All (0.06–0.12 pixels), and sky emission lines in the calibrated science frames lie within 0.57 Å of their laboratory wavelengths. All science slits of the two multi-slit masks are calibrated; the slitmask alignment holes, which the pipeline traces as slits, are not. We close with recommendations for observers and for the observatory.

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1 Introduction

LDSS3-C is the facility imaging spectrograph of the 6.5 m Magellan Clay telescope at Las Campanas Observatory [7]. It combines an 8.3 arcmin diameter field of view with laser-cut slitmasks, three VPH grisms and a single 2048×4096 CCD, and it is widely used for low-resolution optical spectroscopy of faint targets.

PYPEIT [8, 9] is a community-developed Python pipeline that reduces data from more than 40 slit and echelle spectrographs with a common set of algorithms: automatic slit tracing, two-dimensional b-spline sky subtraction [5], optimal extraction [4], and wavelength calibration against archived templates. Adding an instrument amounts to describing its detector, header conventions and default parameters in a spectrograph class; the rest of the pipeline is shared and maintained upstream.

This report documents the LDSS3-C implementation, currently hosted on the `ldss3_multigrating` branch of a fork of PYPEIT [2], and evaluates it on data of every grism. Section 2 presents the instrument and the data sets. Section 3 describes each stage of the reduction and the instrument-specific choices behind it. Section 4 reports the detector characterisation, Section 5 the wavelength-calibration performance and Section 6 example reductions, and Section 7 lists the known limitations. The scripts that produce every number, table and figure in this report are distributed with it [3].

2 Instrument and data

2.1 Detector format

The LDSS3-C CCD is read through two amplifiers, each covering half of the detector in the spatial direction. The data system writes each amplifier to its own FITS file, with a common exposure stem and the suffixes `c1` and `c2` (e.g. `ccd1940c1.fits` and `ccd1940c2.fits`). Each file holds 1024 data columns and 128 overscan columns by 4096 rows (plus overscan rows), with the dispersion along the long axis. After overscan subtraction and trimming, the assembled frame is 4096×2048 pixels in the PYPEIT convention (spectral $\times$ spatial), with amplifier 1 covering spatial pixels 0–1023 and amplifier 2 pixels 1024–2047 (Figure 1). We refer to the boundary between the two halves as the *seam*. The plate scale is 0.189 arcsec/pix.

2.2 Grisms

Table 1 summarises the three grisms as calibrated in this work. The wavelength coverage and dispersion are measured from the arc-lamp solutions of Section 5 for a slit at the centre of the field; in multi-slit mode the coverage shifts with the slit position along the dispersion direction.

Table 1: The LDSS3-C grisms as calibrated here, for the central segment of the 1.0 arcsec longslit: wavelength range of the solution and range covered by identified arc lines, dispersion and central wavelength, arc-line FWHM and resolving power, and the order-blocking filter used in the data of this report.

Grism	Range (Å)	Lines (Å)	Å/pix	λ_c (Å)	FWHM (Å)	R	Filter
VPH-Blue	3734–6497	3890–6418	0.690	5049	3.1	1611	none
VPH-Red	5853–10588	5854–9787	1.176	8149	5.4	1497	OG590
VPH-All	2694–10620	3890–10473	2.029	6411	8.8	730	none

2.3 Data sets

We use five data sets (Table 2). The three longslit sets come from the Las Campanas commissioning archive and include arcs and flats for each grism, and spectrophotometric standards for

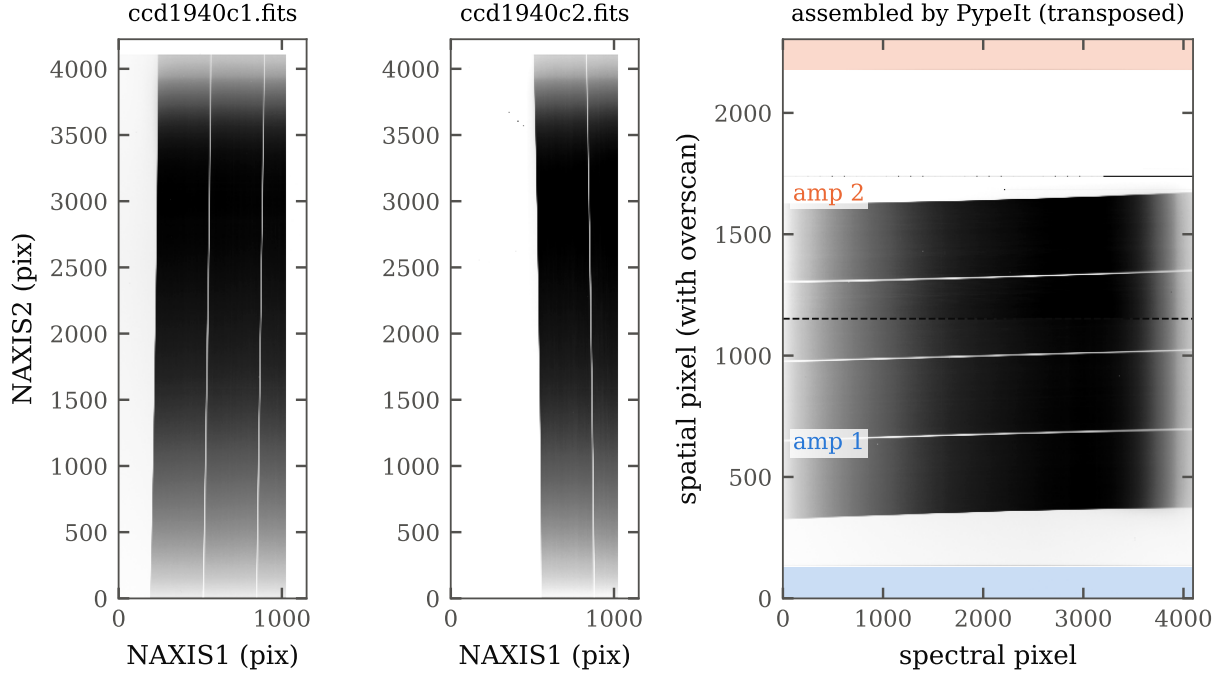


Figure 1: A VPH-Red longslit flat (exposure 1940, 2019 February). Left: the two raw amplifier files as written by the instrument. Right: the frame assembled by PYPEIT, before trimming, with the spectral axis horizontal. Shaded bands mark the overscan regions of amplifiers 1 (blue) and 2 (orange); the dashed line marks the seam between the two data sections.

Table 2: Data sets reduced for this report: number of arc, flat and science/standard exposures used.

Grism	Slit/mask	Filter	Date (UT)	Arcs	Flats	Sci.	Targets
VPH-Blue	1.0_center	Open	2019-02-19 – 02-22	2	6	0	–
VPH-Red	1.0_center	OG590	2019-02-20	3	5	4	GD71
VPH-All	1.0_center	Open	2018-04-03 – 04-04	3	5	4	EG274, Hiltner600
VPH-Blue	ATM3a2_1	Open	2022-11-15 – 11-16	2	7	4	ATM3a2_1
VPH-Red	CDFS01	OG-590	2025-12-22 – 12-23	3	10	3	CDFS01

VPH-Red and VPH-All. The two multi-slit sets are science observations: the VPH-Blue mask ATM3a2_1 (2022 November) and the VPH-Red mask CDFS01 (2025 December). All data were taken in the Fast readout mode with 1×1 binning. Bias and dark frames from all epochs are used only for the detector measurements of Section 4; they are not used in the reductions (Section 3.1).

3 Reduction procedure

The reduction follows the standard PYPEIT multi-slit path [9]. A reduction is set up with `pypeit_setup`, which reads the headers, classifies the frames and groups them into configurations, and run with `run_pypeit`. We describe each stage below, emphasising the choices made for LDSS3-C. The complete set of default parameters is listed in Section A.

3.1 Frame assembly and basic image processing

During setup, PYPEIT ingests only one file per exposure (the c1 file). When a frame is read, the companion amplifier file is located from the file name and the two halves are joined in memory; no pre-merged files are needed. The gain and read noise are assigned per amplifier: the gain

Table 3: Frame classification inferred from the `OBJECT` card. These are heuristics, not header keywords, and must be verified against the observing log.

OBJECT contains	Frame type	Examples from the archive
arc, arcs, henear, hene, lamp, comp	arc, tilt	“arc ATM3a2_1”, “GD71 VPH-RED OG59
flat, quartz, dome, qh, ff	pixel flat, illumination flat, trace	“EG274 - Qh”, “LTT3218 flatQh”
zero, bias	bias	“Zero”, “Bias”
dark	dark	“Dark”
align, thr, field, focus, acq	ignored	“align MASK”
anything else	science or standard	“science ATM3a2_1”, “EG274 - spec”

is read from the `EGAIN` keyword, and the read noise is taken from the published value for the readout mode given by the `SPEED` keyword, not from the `ENOISE` keyword, which we find to be incorrect (Section 4.2).

For each amplifier the overscan is subtracted row by row and the image is trimmed to the data section. Bias frames are not subtracted by default: the bias residuals left after overscan subtraction are below 2 ADU (Figure 2), and small compared with the read noise, and subtracting a combined bias would add noise for little gain. Dark frames are likewise not subtracted; the dark current (Section 4.3) is small and is accounted for in the noise model. Cosmic rays are identified in science frames with the `PYPEIT` L.A.Cosmic implementation (`sigclip` = 5, `objlim` = 2).

Saturation is set by the 16-bit analogue-to-digital converter, not by the full well. The CCD full well is $\sim 205\,000\,e^-$ and the response stays linear to within 1 % up to $\sim 175\,000\,e^-$, but 65535 ADU corresponds to only 108 133 e^- (amplifier 1) and 96 336 e^- (amplifier 2). The saturation threshold is therefore set to $65535 \times \min(g_1, g_2) = 96\,336\,e^-$, with a non-linearity fraction of 0.99, so that saturated pixels are flagged on both amplifiers.

3.2 Frame classification

The `EXPTYPE` keyword does not distinguish reliably between calibration types: arcs and flats are written both with `EXPTYPE` = ‘Flat’ and with `EXPTYPE` = ‘Object’, depending on how the exposure was started, and bias frames carry whatever grism was in the beam. Frames are therefore classified by combining `EXPTYPE` with keywords in the `OBJECT` card (Table 3). Short keywords must appear as separate words (delimited by spaces, underscores or hyphens), so that, e.g., a target named “Arcturus” is not taken for an arc; `flat`, `quartz` and `dome` are matched anywhere in the name, which catches `flatQh`. Frames taken without a grism (acquisition, through-slit and direct images) are discarded. Science frames and standards are separated by exposure time, at 100 s by default.

It must be stressed that the frame type cannot be determined from the header alone. `LDSS3-C` writes no keyword that identifies an exposure as an arc, a flat or a science frame: `EXPTYPE` takes only the values `Object`, `Flat`, `Bias` and `Dark`, and, as noted above, arcs and flats are written with either of the first two. What the pipeline does is therefore a naive inference from the free text of the `OBJECT` card, which the observer is at liberty to write as they please. It reproduces the naming conventions we have encountered in the archive (Table 3) and will fail, silently, on any convention it has not seen. **The classification must always be checked against the observing log before the data are reduced.** The `.pypeit` file lists one row per exposure together with the type assigned to it, and editing that column is the supported way of correcting it.

We inspected the classification of all frames in the configurations used here. Two frames were misclassified: an arc named “Hiltner600 - NeHeAr”, typed as a standard because `nehear` is not among the keywords, and a 1 s test exposure named “Tests”, typed as a standard. The first was retyped and the second removed by hand in the input file, which is the supported way

Table 4: Columns masked by default, in pixels of the trimmed, unbinned frame (spatial axis; amplifier 1 is 0–1023).

Columns	N	Amp.	Columns	N	Amp.
0–11	12	1	1602–1606	5	2
443	1	1	1635–1640	6	2
608	1	1	1688–1690	3	2
1413	1	2	1695	1	2
1492–1494	3	2	1999–2000	2	2
1549–1551	3	2	2034–2047	14	2

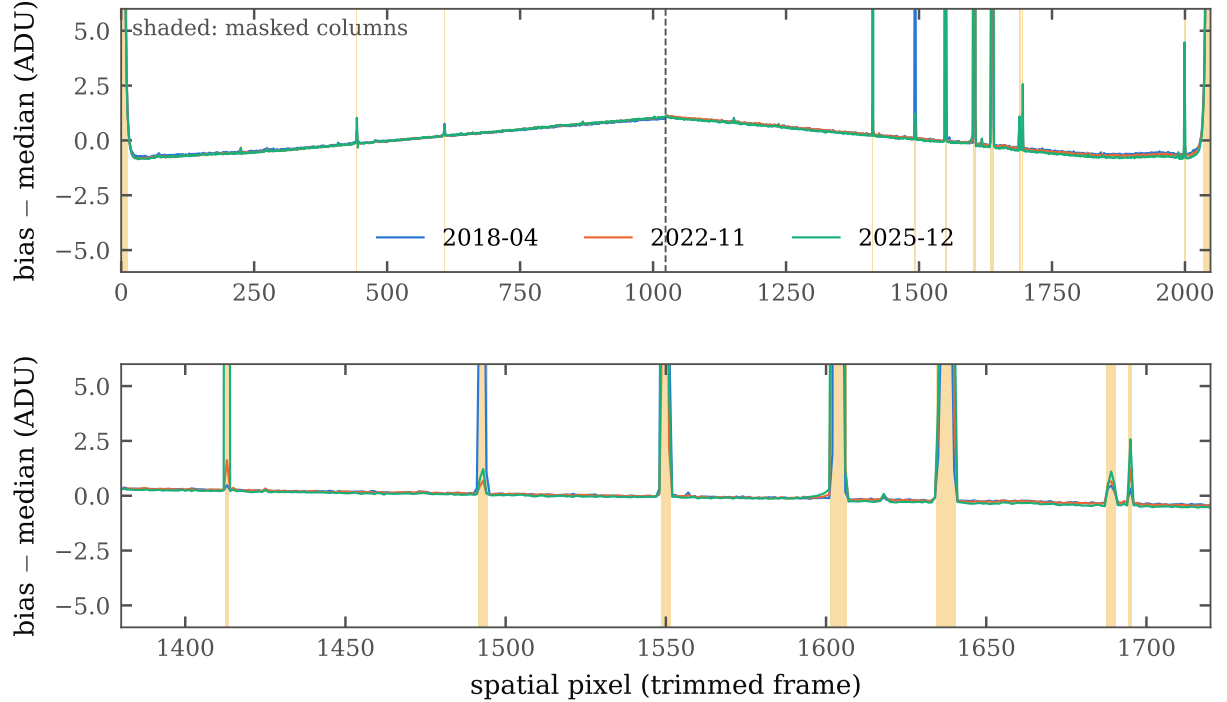


Figure 2: Column profile of combined, overscan-subtracted bias frames at three epochs, relative to the median of amplifier 1. Shaded columns are those masked by default. The lower panel is an enlargement of the region with most bad columns. The dashed line marks the seam.

of overriding the automatic classification.

A configuration is defined by the grism, the slitmask (`APERTURE`), the binning and the order-blocking filter. The filter is included because it changes the spectral shape of the flat field; frames taken with different filters are never calibrated together automatically (see Section 3.5).

3.3 Bad-pixel mask

The default bad-pixel mask flags twelve column ranges (Table 4), including the outer edge columns of each amplifier. They were identified from the column profile of combined biases and confirmed at 5 epochs between 2018 and 2025 (Figure 2). Flagging columns that deviate from the local median by more than 8 times the column-to-column scatter, every masked range is detected in at least four of the five epochs. The only other columns detected that often are 12–17 and 2028–2033, adjacent to the masked outer edges of the detector; they lie in the edge roll-off and are not inside any slit in the data analysed here. Apart from these columns, the overscan-subtracted bias is flat to within 2 ADU, with a smooth gradient of about 1.5 ADU towards the seam.

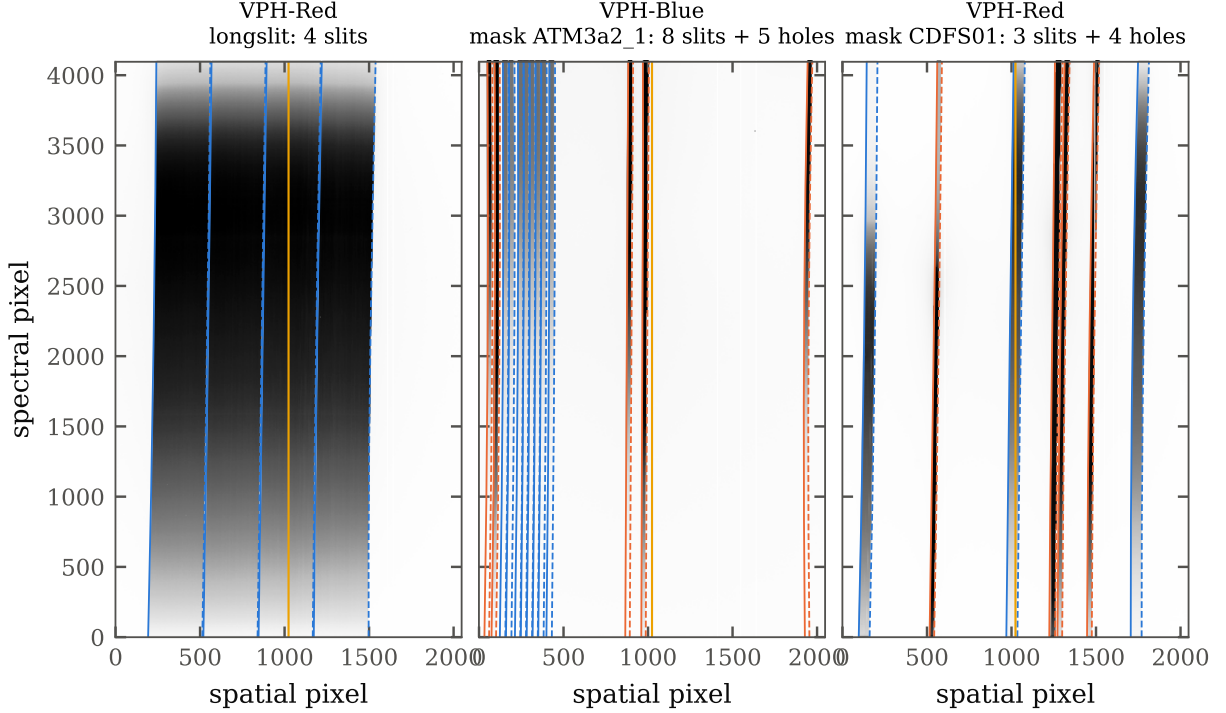


Figure 3: Traced slit edges (solid: left edge; dashed: right edge) over the combined trace flat, for the VPH-Red longslit and the two multi-slit masks. Slits flagged by the pipeline are shown in orange. The yellow line marks the seam.

3.4 Slit tracing

Slit edges are found in the combined trace flat from the spatial gradient of the illumination and followed along the dispersion direction. For LDSS3-C the edge-detection threshold is raised to `edge_thresh` = 20: at lower thresholds, bad columns and the detector edges produce spurious narrow slits. Slits shorter than 2 arcsec (`minimum_slit_length`) are rejected; the narrowest real slitlets we have encountered are ~ 5 arcsec long. In multi-slit mode, missing edges are predicted from a principal-component model of the other slits; for the longslit masks (`APERTURE` containing `center` or `long`) the prediction uses the nearest traced slit instead.

The LDSS3-C longslits are not continuous: the slit is cut in segments separated by narrow bridges that hold the mask together. PYPEIT traces each segment as a separate slit (Figure 3, left). For the 1.0 arcsec `center` slit we find 4 segments of 60 arcsec each, separated by bridges of ~ 6 pix. This has no practical consequence for the reduction, but an object that falls on a bridge is lost, and each segment receives its own wavelength solution.

Figure 3 also shows the two multi-slit masks, which we compared with their design files. ATM3a2_1 has 8 slits of 1×6 arcsec and 6 alignment holes of 5 arcsec diameter; CDFS01 has 3 slits of 1×12 arcsec and 9 holes. All science slits were traced (8 and 3). The alignment holes are also traced, as short slits: 5 in ATM3a2_1 and 4 in CDFS01, where several holes share the same spatial position. The holes are wider than the slits in the dispersion direction, so they are over-exposed in flats and arcs and their arc lines are broad; they cannot be calibrated and must be ignored (Section 5.1). A slit whose edge falls on a masked bad column cannot be traced there and may be lost; this did not occur for the two masks studied, but should be considered in mask design.

3.5 Flat fielding

The flat-field frames (internal quartz lamp; Qh in the LDSS3 nomenclature) are combined and decomposed, slit by slit, into a spectral response, a spatial illumination profile and a pixel-to-

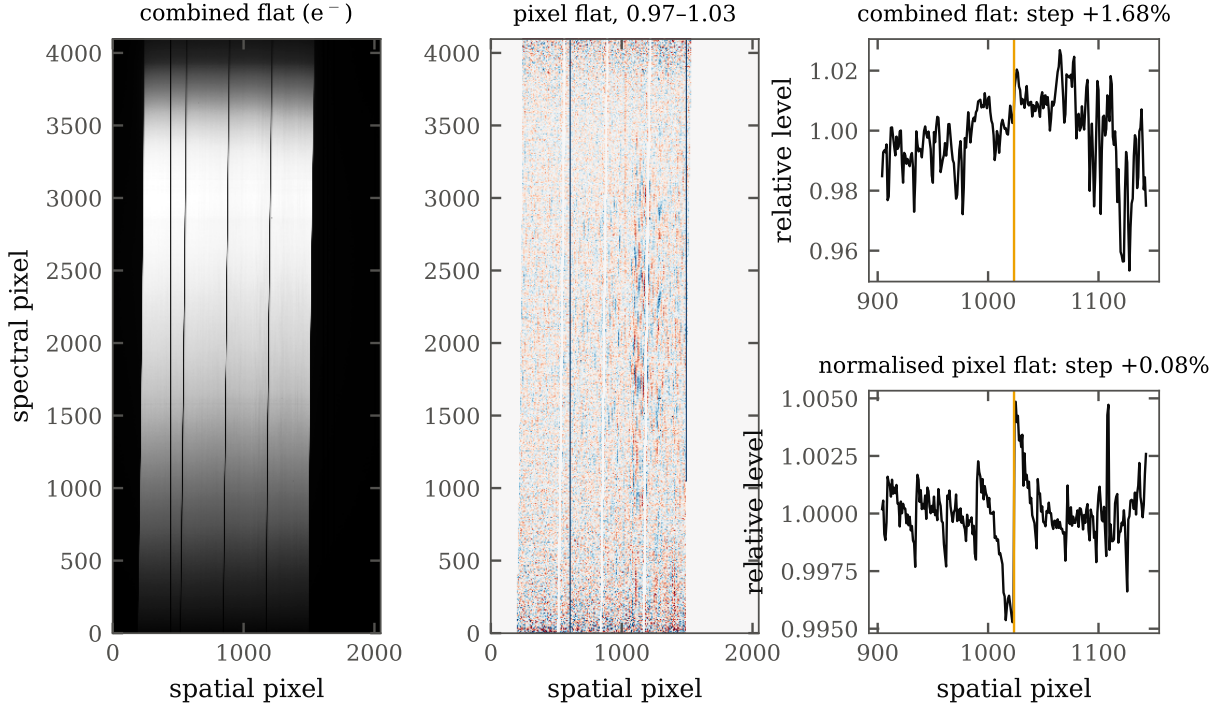


Figure 4: Flat field for the VPH-Red longslit. Left: combined flat (in electrons). Centre: normalised pixel-to-pixel flat. Right: median spatial profile around the seam before (top) and after (bottom) normalisation; the quoted step is measured by extrapolating a linear fit from each side to the seam.

pixel sensitivity map (Figure 4). The pixel flat corrects for pixel-scale sensitivity variations with an RMS of 0.7% in the VPH-Red longslit data.

Because the two amplifiers have slightly different gains, a flat shows a step at the seam. Measured inside the longslit segment that crosses the seam, it is 0.8–2.4% in the combined flats and 0.01–0.08% after normalisation, confirming that the two halves are joined and scaled consistently. None of the multi-slit slits spans enough of the seam for the measurement.

The last step of the illumination model is a fine correction of the spatial profile of each slit, fitted in bins of wavelength. It is the step most sensitive to the quality of the flat, and its QA plots (QA/PNGs/Spatillum_FineCorr_*) should be inspected. In the data analysed here, inspected slit by slit, the correction amounts to 1–2% across the slit in the longslit segments and in the science slits of ATM3a2_1, and follows the data smoothly. In the science slits of CDFS01, whose flats were taken without the blocking filter (see below), it reaches 3–4% and is fitted to visibly noisier profiles. It fails in two recognisable ways, both in alignment holes. Where the flat is saturated, only the unsaturated part of the aperture enters the fit and the profile is extrapolated over the rest; this occurs in the holes of CDFS01, whose flats reach the ADC ceiling. In ATM3a2_1, one wavelength bin of one hole shows a spurious gradient of $\sim 3\%$.

The VPH-Red flats of CDFS01 were taken without the OG-590 order-blocking filter used for the arcs and science frames. Such frames form a separate configuration and are not associated automatically; for this report we combined them by hand. This is adequate for the pixel-to-pixel correction and for slit tracing, but the spectral shape of an unfiltered flat differs from that of the science data below $\sim 6000 \text{ \AA}$, and the extra light is what saturates the alignment holes. Flats should be taken with the same filter as the science.

3.6 Wavelength calibration

Arc spectra are extracted along the centre of each slit, and the lines are detected above a threshold of `sigdetect` = 6 with an expected FWHM of 6 pix. The wavelength solution is

found with the `full_template` method: the arc spectrum of each slit is cross-correlated with an archived template for the same grism (Figure 7), allowing for a shift and a stretch, and the template’s identified lines are transferred to the observed spectrum within a tolerance of 2.5 pix. A Legendre polynomial of order `n_final` = 5 is then fitted to the pixel positions of the identified lines, with iterative rejection of outliers; solutions with an RMS larger than half the line FWHM are rejected. The line lists are those distributed with PYPEIT for the HeI, NeI and ArI lamps, in vacuum wavelengths [6].

The templates are HeNeAr spectra of the 1 arcsec longslit, one per grism, whose wavelength scales were checked against the reference arc identifications distributed by Las Campanas Observatory for each grism. They cover the full detector. Because the template is matched to each slit, the same template serves slits at any position on a multi-slit mask.

The wavelength solution is extended to two dimensions by tracing the arc lines along each slit and fitting the line tilts with a two-dimensional polynomial of order 6 in both directions.

3.7 Sky subtraction and extraction

The sky is modelled in two dimensions, slit by slit, with a b-spline in the wavelength–spatial plane [5] with a break-point spacing of 0.8 pixels. Objects are found in the sky-subtracted image and extracted with both a boxcar and an optimal extraction [4]; in the second pass the sky is re-fitted jointly with the object profiles. For standard stars the global sky subtraction is skipped. On multi-slit data we recommend excluding the outer parts of each slitlet from the object search to avoid spurious detections at the edges (`find_trim_edge`; Section C).

3.8 Flexure correction

Arcs are usually taken at a different telescope position from the science frames, so a residual shift in the spectral direction is expected. It is measured by cross-correlating the extracted sky spectrum of each object with an archived sky spectrum and applied to the wavelength scale (`spec_method` = `boxcar`). For the science slits in this study the measured shifts range from -2.5 to +0.1 pixels. The largest, about -2.4 pix, is common to all slits of CDFS01, for arcs taken during the night close to the science exposures; it is confirmed independently by the sky lines (Section 5.2). Standards are not corrected. Flexure should be checked in the QA plots (QA/PNGs/*spec_flex_corr*).

3.9 Flux calibration

Sensitivity functions are computed with `pypeit_sensfunc` using the IR algorithm, which fits the instrument response jointly with a telluric model, with a polynomial of order 7. The flux calibration has not been validated and is not assessed in this report.

4 Detector characterisation

4.1 Header keywords

All 293 exposures surveyed here, from 2018 April to 2025 December, carry the same detector keywords: `SPEED` = `'Fast'`, `EGAIN` = 1.65 and 1.47 e⁻ ADU⁻¹, and `ENOISE` = 4.67 and 5.06 e⁻ for amplifiers 1 and 2. The `ENOISE` values are close to the published Slow-mode values, not to the Fast-mode ones. The published read noise is 4.4 e⁻ and 5.3 e⁻ (Slow), 7.0 e⁻ and 7.2 e⁻ (Fast) and ~10 e⁻ (Turbo) for the two amplifiers.

Table 5: Read noise (e^-) per epoch, from bias-pair differences (mean $\pm$ standard deviation over pairs) and from the overscan scatter, for the gains in the **EGAIN** keyword. The published Fast-mode values are 7.0 and 7.2 e^- ; the **ENOISE** keyword gives 4.67 and 5.06 e^- .

Epoch	N_{bias}	Amplifier 1		Amplifier 2	
		pairs	overscan	pairs	overscan
2018 Apr	5	6.43 ± 0.10	6.21	6.82 ± 0.12	6.56
2019 Dec	5	6.74 ± 0.26	6.62	7.13 ± 0.26	6.94
2020 Nov	5	6.61 ± 0.84	6.72	7.03 ± 0.81	7.11
2022 Nov	10	6.24 ± 0.39	6.09	6.67 ± 0.42	6.46
2025 Dec	13	6.17 ± 0.26	6.17	6.62 ± 0.28	6.54
Mean		6.44 ± 0.11		6.85 ± 0.10	

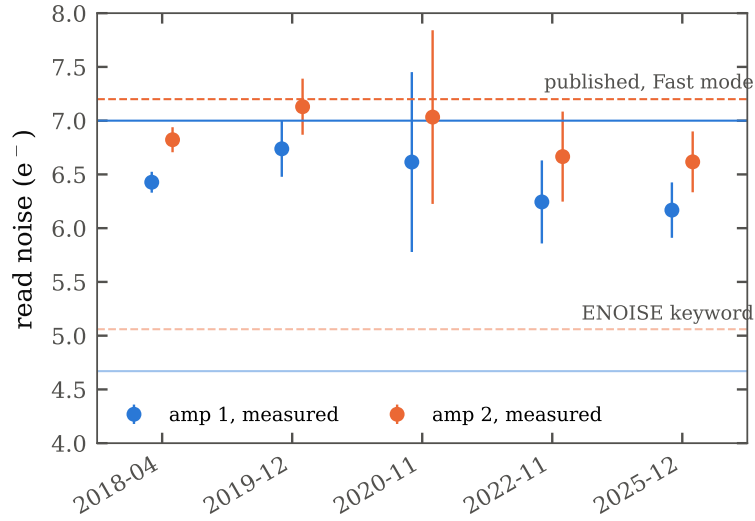


Figure 5: Read noise measured from bias pairs at each epoch. Horizontal lines are the published Fast-mode values (upper) and the **ENOISE** keyword values (lower) for amplifier 1 (solid) and 2 (dashed).

4.2 Read noise

The read noise was measured at 5 epochs from the differences of consecutive bias frames, $\sigma_{\text{RN}} = \sigma(B_i - B_{i+1})/\sqrt{2}$, in the central region of each amplifier after overscan subtraction, with iterative 4σ clipping. It was also measured independently from the scatter in the overscan. The results are listed in Table 5 and shown in Figure 5. The mean over epochs is 6.44 ± 0.11 and 6.85 ± 0.10 e^- , consistent with the published Fast values and 35–38 % larger than the **ENOISE** keyword.

The read noise enters the variance of every pixel, so the choice matters most where the sky is faint, in the blue and between the sky lines. We tested it with the distribution of $\chi = (d - s - o)\sqrt{w}$ over the sky-dominated pixels of the science exposures (excluding the alignment holes), where d , s and o are the data, the sky model and the object model and w is the inverse variance. With the published read noise the distribution has a standard deviation of 0.98 (0.95–1.05 for individual exposures). Replacing the read noise with the **ENOISE** values gives 1.20: the uncertainties would be underestimated by 20 % (Figure 6). Standard stars are excluded from this test because they are reduced without a global sky model. The pipeline therefore takes the read noise from the readout mode and falls back on **ENOISE**, with a warning, only for an unrecognised mode.

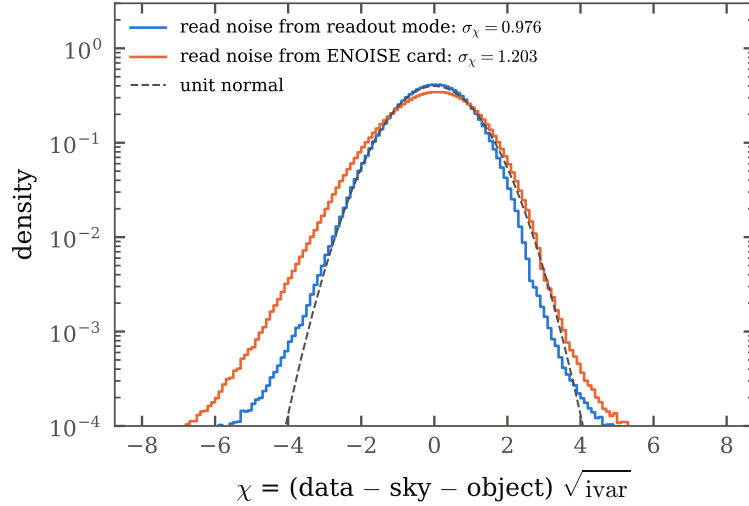


Figure 6: Distribution of the normalised residuals χ over the good pixels of the science frames in Table 2, for the variance computed with the published read noise (as used by the pipeline) and with the **ENOISE** header values. The dashed line is a unit normal.

Table 6: Wavelength-calibration performance. For each data set and calibration group: slits with a valid solution out of the science slits traced (alignment holes excluded); and, as medians over those slits, the number of lines in the fit, the RMS residual in Å, pixels and km s^{−1}, the FWHM of the arc lines and the resolving power $R = \lambda_c/\text{FWHM}$ at the central wavelength for the 1 arcsec slit.

Data set	Group	Slits	N_{lines}	RMS (Å)	RMS (pix)	RMS (km/s)	FWHM (Å)	R
VPH-Blue, 1.0" longslit	0	4/4	31	0.045	0.06	3	3.2	1576
VPH-Blue, 1.0" longslit	1	4/4	26	0.042	0.06	2	3.2	1580
VPH-Red, 1.0" longslit	0	4/4	46	0.147	0.12	5	5.5	1480
VPH-All, 1.0" longslit	0	4/4	66	0.245	0.12	11	8.7	736
VPH-All, 1.0" longslit	1	4/4	60	0.218	0.11	10	8.7	738
VPH-Blue, mask ATM3a2_1	0	8/8	41	0.077	0.11	5	3.2	1573
VPH-Red, mask CDFS01	0	3/3	52	0.082	0.07	3	6.2	1321

4.3 Dark current

The dark current, measured from dark exposures of 120–900 s bracketed by biases (sigma-clipped mean of the difference), is 7.2–25.8 e[−] pix^{−1} h^{−1}, with amplifier 2 consistently about twice amplifier 1. The pipeline adopts 25 e[−] pix^{−1} h^{−1} in its noise model. For a 20 min exposure this amounts to 2–9 e[−] per pixel, well below the sky level in any spectroscopic exposure but not negligible compared with the read noise, so the dark current is included in the variance but not subtracted.

5 Wavelength-calibration performance

Figure 7 shows the three arc templates with the lines used in the solution, and Figure 8 shows the residuals of the longslit solutions. Table 6 summarises the performance for every data set. For the multi-slit masks, Figure 9 shows the dispersion, number of lines and RMS of every slit.

VPH-Blue. The solution is constrained by HeI, NeI and ArI lines from 3890–6418 Å, with 26–44 lines per slit and a median RMS of 0.04–0.08 Å per data set (~ 0.1 pix; 0.04–0.17 Å for individual slits). The blue half of the range has few, faint lines (Figure 7), so the solution there relies on HeI 3889, 4472, 4714 and 5017 Å and ArI 4159 and 4301 Å; the Ne lines cluster redward

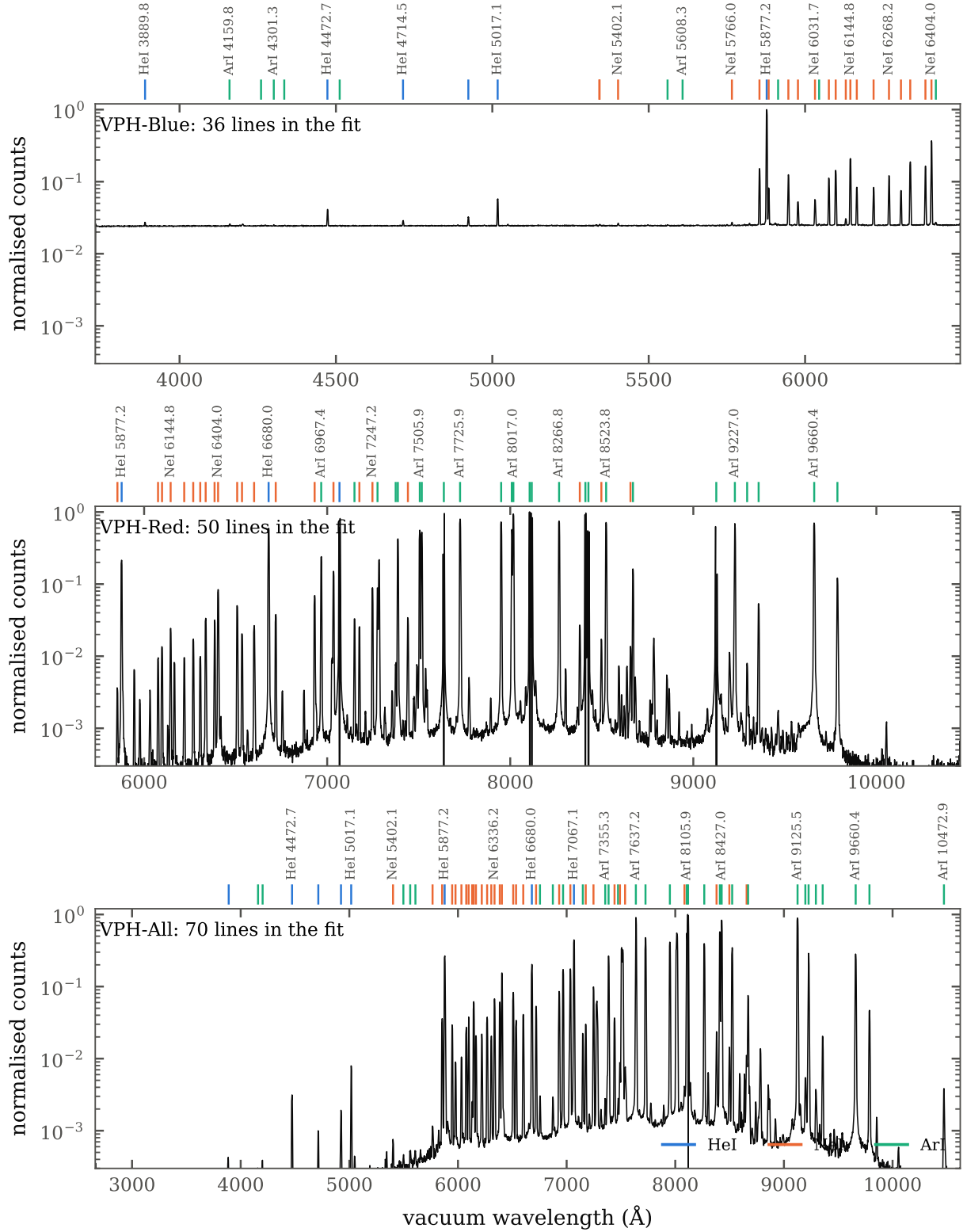


Figure 7: Archived HeNeAr arc templates for the three gratings, with the lines used in the wavelength solution of the longslit data marked by element (HeI blue, NeI orange, ArI green). The brightest lines are labelled with their vacuum wavelength in Å.

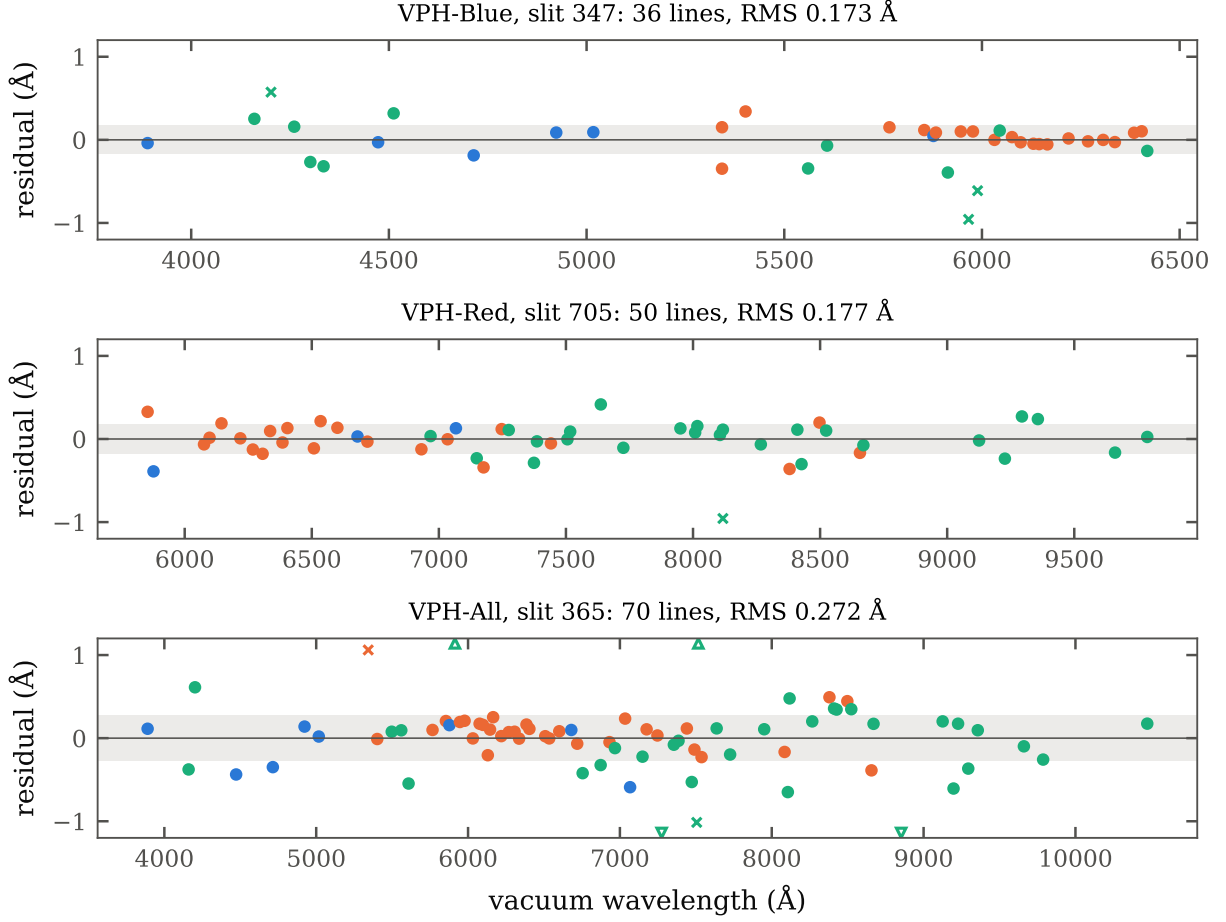


Figure 8: Residuals of the wavelength solutions for the longslit data, for the slit segment with the most identified lines. Filled symbols are lines used in the fit, crosses are rejected lines; the shaded band is $\pm$ RMS.

of 5800 Å. In ATM3a2_1 the eight science slits give dispersions within 0.5% of each other and coverage that shifts by up to 110 Å with slit position.

VPH-Red. Lines from 5854–9787 Å constrain the solution, with 36–62 lines per slit and an RMS of 0.05–0.18 Å. The reddest $\sim$ 800 Å of the detector, beyond the last strong ArI line, are extrapolated. The sky lines show that the extrapolation is adequate: the 184 sky-line measurements redward of 9800 Å give a median offset of +0.42 Å (scatter 0.91 Å). Slits placed so that their spectra start blueward of the first arc line are also extrapolated at the blue end; in CDFS01 the [OI] 5577 Å sky line is found 1.5 Å too blue there, in a range the OG-590 filter blocks anyway.

VPH-All. The coverage of VPH-All is limited by the arc lamps, not by the detector: lines are identified from 3890–10 473 Å, and the first $\sim$ 600 pix (below 3890 Å) are extrapolated, where the grism has no throughput and no signal is recorded. With 60–70 lines per slit the RMS is 0.19–0.27 Å, or 0.1 pix.

Continuity across the seam. Extrapolating the wavelength image from each side to the seam, the median discontinuity over the rows of a slit is at most 0.009 Å, and the largest in any single row is 0.10 Å (0.08 pix, in the CDFS01 slit that crosses the seam), well below the dispersion.

Table 7: Wavelength zero point of the science frames, from Gaussian fits to isolated sky emission lines ([OI] and OH) in the final, flexure-corrected wavelength frame. For each data set: number of slit-exposure pairs, median number of lines per slit, and the median and largest per-slit offset (measured – reference).

Data set	Slits×exp.	N_{lines}	Median (Å)	Max. $ \Delta $ (Å)	Max. $ \Delta $ (pix)
VPH-Red, 1.0" longslit	13	66	+0.07	0.15	0.12
VPH-All, 1.0" longslit	7	33	-0.09	0.57	0.28
VPH-Blue, mask ATM3a2_1	32	5	-0.12	0.28	0.41

5.1 Alignment holes

The alignment holes of a slitmask are traced as short slits (Section 3.4). Because they are much wider than a slit in the dispersion direction, their arc lines are broad and flat-topped, often saturated, and the template match fails or produces a wrong solution (Figure 9). Such a solution can still have a small RMS if it matches a few lines, so the dispersion and the line width, rather than the RMS, identify it: in both masks every hole has an arc-line FWHM more than 1.5 times that of the slits or a failed solution, and no science slit does. `ldss3_qa.py` flags these apertures automatically. Objects found in them are the alignment stars.

5.2 Zero point from the sky lines

The arc solutions are applied to science frames taken at other times and telescope positions, corrected for flexure (Section 3.8). To test the final wavelength scale independently of the arcs, we fitted Gaussian profiles to isolated sky emission lines in the extracted sky spectrum of every science slit: the [OI] lines at 5577, 6300 and 6364 Å and the brightest OH lines of OH_LDSS3_vac with no other listed line within 8 Å (after removing the heliocentric correction applied by the pipeline). Fits with widths inconsistent with the instrumental profile, which are blends, were rejected. Table 7 lists the results. The median offset is at most 0.12 Å in absolute value for every data set and the largest per-slit offset is 0.57 Å (0.4 pix). The line-to-line scatter, 0.5–1 Å, is dominated by unresolved blends in the reference list at this resolution rather than by the solution.

Sky-line calibration. For long VPH-Red exposures, the OH sky lines provide an alternative calibration that is recorded through the same optical path and at the same telescope position as the science data. We provide a sky-line template for VPH-Red and a list of 300 OH lines between 6000 and 10 400 Å (OH_LDSS3_vac; Figure 10), derived from the UVES sky atlas [1] and converted to vacuum. Unresolved groups within 3 Å were merged to their flux-weighted centroid. The parameters to enable this mode are given in Section C; the flexure correction must then be disabled, since the solution is already in the frame of the observation. This mode has not been validated end to end and is not used in this report.

6 Example reductions

Figure 11 shows, for each data set in which a source was detected, the sky-subtracted two-dimensional spectrum and the optimally extracted one-dimensional spectrum of the object with the highest signal-to-noise ratio, excluding alignment stars. The longslit standards reach $S/N > 100$ per pixel. The multi-slit targets are faint (knots of a lensed arc in ATM3a2_1), with $S/N \sim 1$ per pixel in a single exposure, and are shown in 8-pixel bins. The VPH-Blue longslit and the VPH-Red mask CDFS01 yielded no detected source and are not considered here; their arc and flat calibration is reported in Section 5 as for the other data sets.

The sky subtraction was assessed with the same χ statistic as in Section 4.2, separately for the 10 % of sky-dominated pixels with the brightest sky and for the rest. In the longslit, σ_χ is

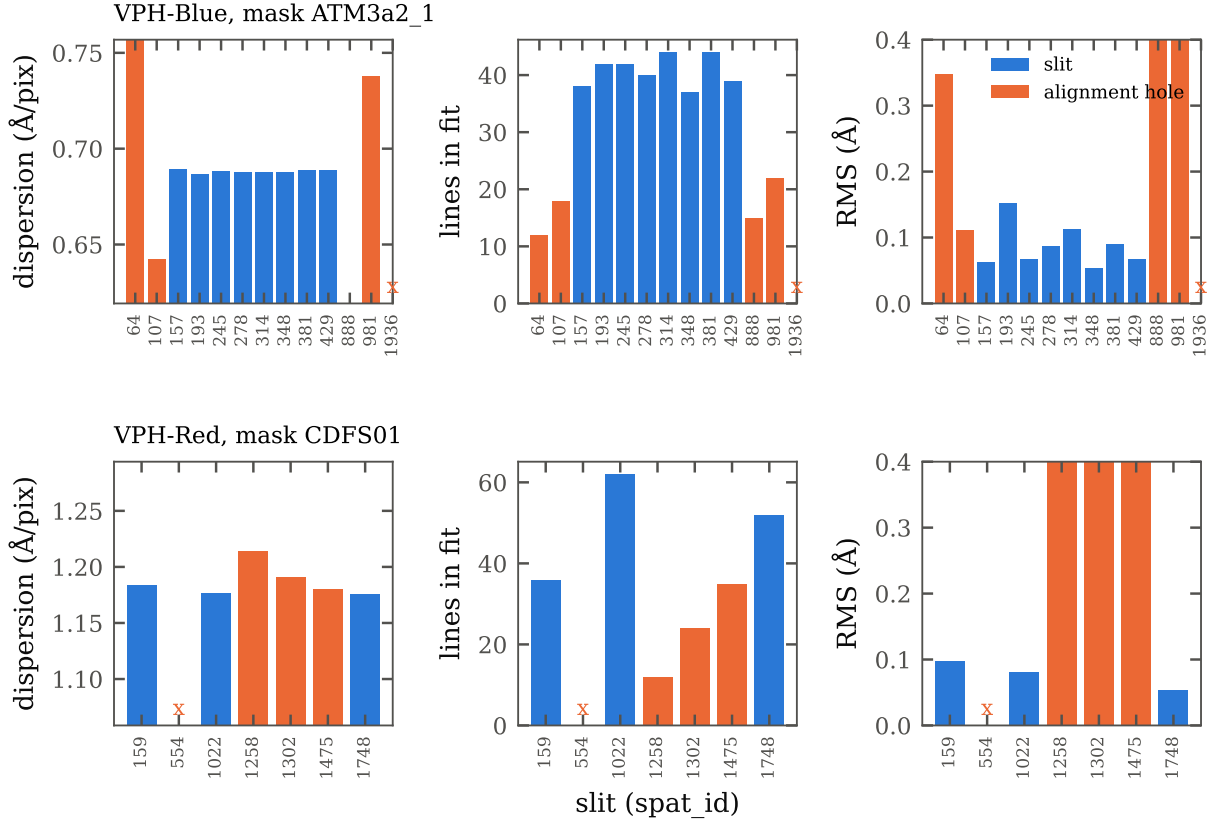


Figure 9: Per-slit dispersion, number of lines used and fit RMS for the two multi-slit masks. Slits whose dispersion deviates by more than 5 % from the median are shown in orange.

0.95 and 1.00 respectively: sky subtraction is noise-limited even on the brightest lines. In the multi-slit mask ATM3a2_1 it is 0.96 and 1.10, and the distribution on bright lines has extended tails: sky-line residuals exceed the noise by 10–20 %. These residuals, visible as spikes at the OH lines in the faint spectra of Figure 11, are typical of short slits (6–12 arcsec), where the sky model has less spatial leverage, and should be considered when interpreting features at bright sky lines.

7 Known limitations and future work

- Only 1×1 binning has been tested. Other binnings produce a warning; the data and overscan sections are then read from the header and should be checked.
- Alignment holes are traced and reduced as slits; they are flagged by `ldss3_qa.py` but not excluded by the pipeline (Section 5.1). The mask design file could be used to identify them.
- Wavelengths are extrapolated beyond the reddest arc line of VPH-Red ($\sim 9790 \text{ \AA}$; validated with sky lines to $0.4 \text{ \AA}$), blueward of the first line in slits whose coverage extends there, and blueward of $3890 \text{ \AA}$ with VPH-All, where there is no signal.
- VPH-All has been tested in longslit mode only. The VPH-Red multi-slit test used flats taken without the blocking filter.
- Sky-line residuals in short multi-slit slits exceed the noise by 10–20 % (Section 6).
- Flux calibration has not been validated.
- The sky-line calibration mode has not been validated end to end.
- The implementation lives in a fork of `PYPEIT`; we plan to contribute it to the main `PYPEIT` repository, after which it will be installed with the standard `PYPEIT` distribution.

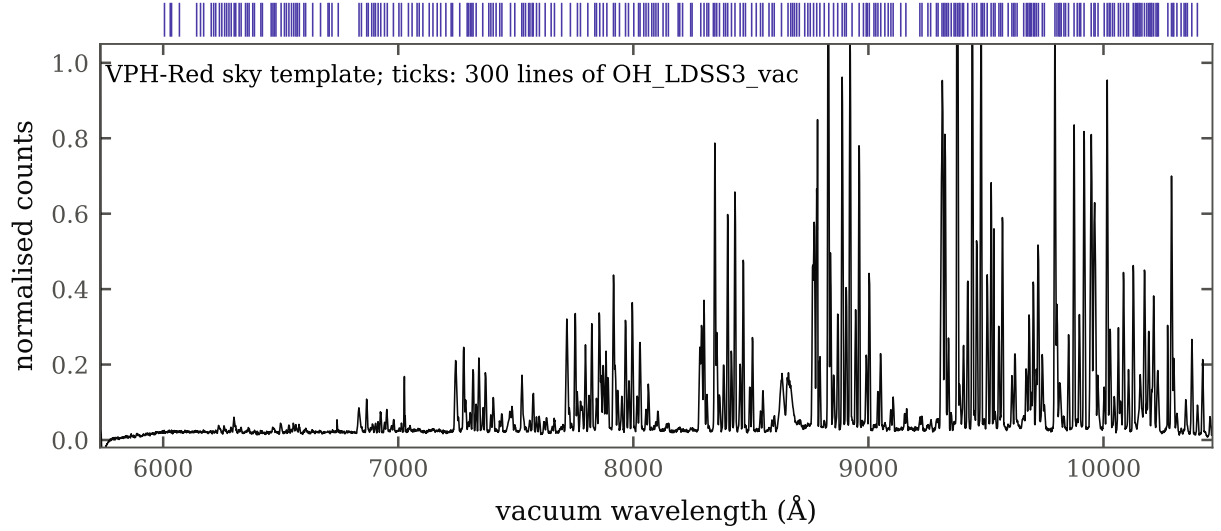


Figure 10: VPH-Red sky template with the positions of the OH_LDSS3_vac lines.

A Default parameters

Table 8 lists the parameters that the LDSS3-C spectrograph class changes from the PYPEIT defaults. The complete parameter set of any reduction is written to the `.par` file in the reduction directory.

Table 8: Parameters set by the LDSS3-C spectrograph class that differ from the PYPEIT defaults (default_pypeit_par). Grism-dependent settings (config_specific_par) are the wavelength template (reid_arxiv), method = full_template, lamps = HeI, NeI, ArI, and fwhm = 6 binned pixels; for longslit masks sync_predict = nearest.

Parameter group	Parameter	Value
rdx	spectrograph	magellan_ldss3
darkframe	exprng	1, None
arcframe	exprng	None, 60
tiltframe	exprng	None, 60
standardframe	exprng	None, 100
flatfield	saturated_slits	mask
wavelengths	sigdetect	6.0
wavelengths	fwhm	5.0
wavelengths	rms_thresh_frac_fwhm	0.5
wavelengths	match_toler	2.5
wavelengths	n_first	3
wavelengths	n_final	5
slitedges	minimum_slit_length	2.0
tilts	spat_order	6
tilts	spec_order	6
scienceframe	exprng	100, None
scienceframe / process	sigclip	5.0
scienceframe / process	objlim	2.0
reduce / skysub	bspline_spacing	0.8
reduce / skysub	global_sky_std	False
flexure	spec_method	boxcar
sensfunc	algorithm	IR

Parameter group	Parameter	Value
sensfunc	polyorder	7
sensfunc / IR	telgridfile	TellPCA_3000_26000_R15000.fits

B Header keywords

Table 9: Header keywords read by the pipeline.

Keyword	Use
OBJECT, EXPTYPE	frame classification, target name
GRISM, APERTURE, FILTER, BINNING	configuration
EXPTIME, AIRMASS, RA, DEC	metadata
UT-DATE, UT-TIME	observation time (MJD)
OPAMP, SPEED, EGAIN	amplifier, readout mode, gain
DATASEC, BIASSEC	data and overscan sections

C Quick start

Install the fork in a fresh environment (Python 3.12):

```
conda create -n ldss3 python=3.12 pip && conda activate ldss3
conda install -c conda-forge pyqt=6
git clone -b ldss3_multigrating https://github.com/joaquinhernandezg/PypeIt.git
cd PypeIt && pip install -e .
```

Inspect the raw data, write the input file for one configuration, check its **frametype** column against the observing log — the types are inferred from the text of the OBJECT card and cannot be taken on trust (Section 3.2) — and reduce:

```
pypeit_setup -s magellan_ldss3 -r /path/to/raw
pypeit_setup -s magellan_ldss3 -r /path/to/raw -c A
cd magellan_ldss3_A && run_pypeit magellan_ldss3_A.pypeit
```

Useful additions to the parameter block of the **.pypeit** file:

```
[reduce]
[[findobj]]
    find_trim_edge = 100, 100    # multi-slit: ignore slit edges
[calibrations]
[[standardframe]]
    exprng = None, 300          # standards up to 300 s
[scienceframe]
    exprng = 300, None
```

Sky-line wavelength calibration for VPH-Red (set the **frametype** of the science frames to **arc,tilt,science**):

```
[calibrations]
[[wavelengths]]
    lamps = OH_LDSS3_vac
    method = full_template
    reid_arxiv = magellan_ldss3_vph_red_sky.fits
[[arcframe]]
    exprng = 0, None
```

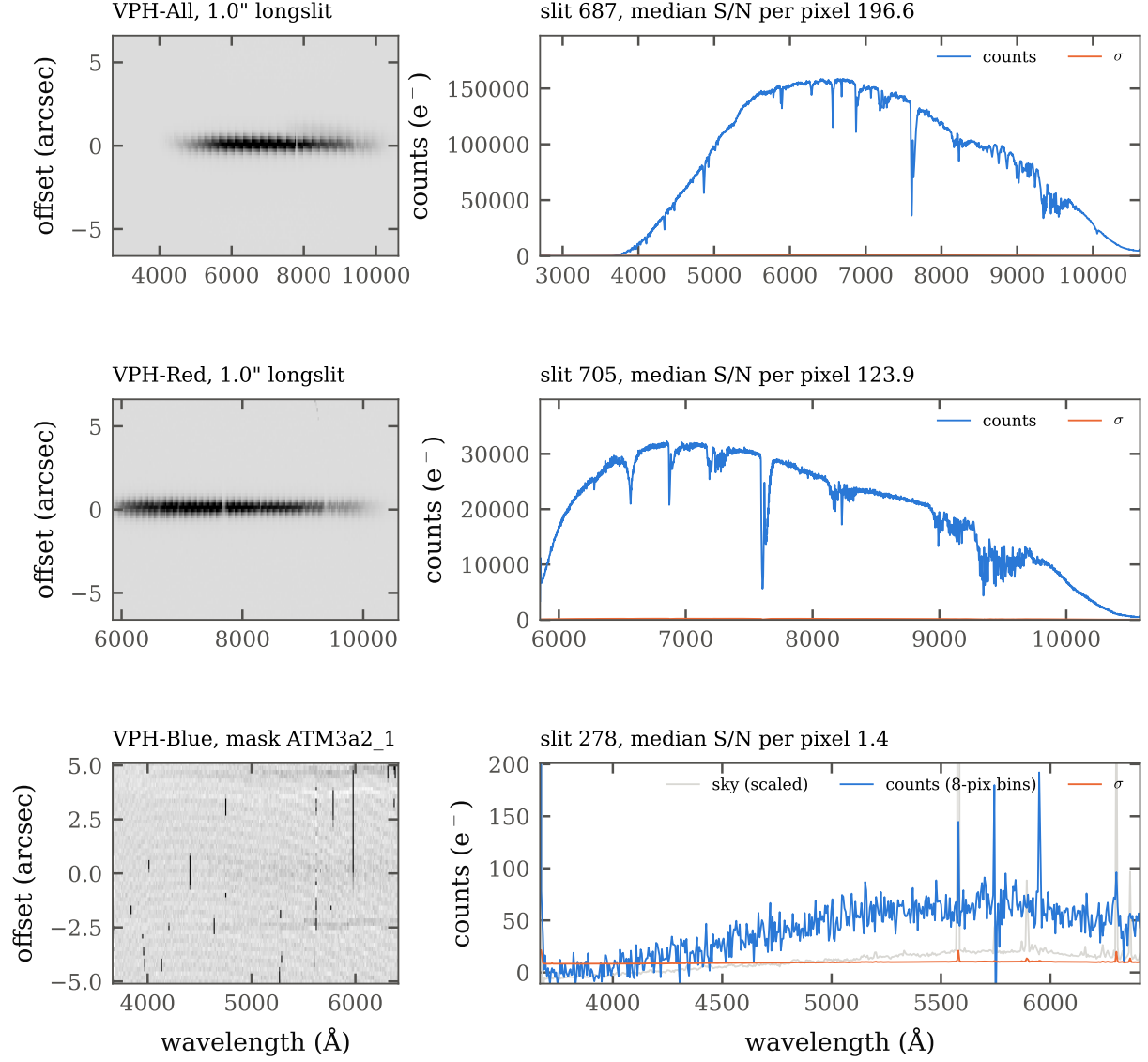


Figure 11: Examples of reduced data. Left: sky-subtracted two-dimensional spectrum around the object, with wavelength increasing to the right. Right: optimally extracted spectrum (blue) and its uncertainty (orange), in electrons, before flux calibration.

```
[[tiltframe]]
    exprng = 0, None
[flexure]
    spec_method = skip
```

Quality assessment and visual inspection:

```
python ldss3-qa/ldss3_qa.py magellan_ldss3_A --tar
pypeit_chk_wavecalib Calibrations/WaveCalib_*.fits
pypeit_chk_edges Calibrations/Edges_*.fits.gz
pypeit_show_2dspec Science/spec2d_*.fits
```

D Reproducing this report

The `report/` directory of Hernández [3] contains the input files of every reduction (`pypeit_files/`) and the scripts that produce all measurements, tables and figures: `prepare_pypeit_files.py` and `run_reductions.sh` (reductions), `measure_detector.py` (Section 4), `measure_calibs.py` (Sections 3, 5 and 6), `make_tables.py` and `make_figures.py`. Running `make` in that directory rebuilds the document. The reductions used PYPEIT version 2.0.1.dev1241+gf3a1f1d27.d20260922.

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- [9] J. Xavier Prochaska, Joseph F. Hennawi, Kyle B. Westfall, Ryan J. Cooke, Feige Wang, Tiffany Hsyu, Frederick B. Davies, Emanuele Paolo Farina, and Debora Pelliccia. PypeIt: The Python spectroscopic data reduction pipeline. *Journal of Open Source Software*, 5(56): 2308, 2020. doi: 10.21105/joss.02308.